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Data-driven porosity monitoring in aluminum laser welding: integration of high-speed imaging and machine learning

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Abstract

Porosity formation remains a critical challenge in laser welding of aluminum alloys (AAs), where unstable keyhole dynamics can lead to internal defects and compromise joint integrity. This study presents a data-driven framework as a foundational step toward real-time porosity monitoring in overlap laser welding of AA6061. A high-speed camera, mounted coaxially on the laser head, continuously records top-view images of the weld pool during the process. These images are analyzed to predict the presence of porosity exceeding 1.5 mm in diameter, a threshold defined by industrial quality standards. To develop and validate the predictive model, two representative 6 mm weld segments, one

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porous and one defect-free, were identified using post-weld X-ray imaging. Six key geometrical features of the melt pool (length, width, area, aspect ratio, eccentricity, and solidity) were extracted from the high-speed images and used as input for training a Random Forest classifier. The model achieved an accuracy of approximately 80% in distinguishing between porous and non-porous weld regions. These results highlight the feasibility of integrating high-speed visual monitoring with machine learning for non-destructive, in-process porosity detection in laser welding applications.

Keywords

Aluminum overlap welding; Feature recognition; Machin learning; Smart manufacturing; Data-driven approach

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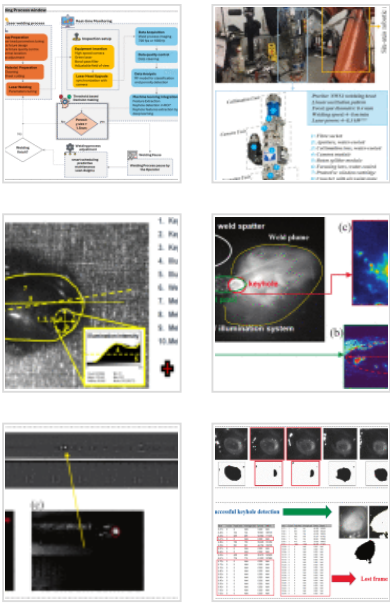
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1. Introduction

The demand for lightweight and energy-efficient vehicles has led to the widespread adoption of aluminum alloys (AAs) in the automotive industry, owing to their low density, corrosion resistance, high strength-to-weight ratio, and recyclability [1]. Among the various joining techniques, laser welding has gained significant attention due to its high-power density, narrow fusion zone, and adaptability to complex geometries [2]. However, laser welding of aluminum remains a challenging process. Properties such as high thermal conductivity, reflectivity, and low viscosity increase the complexity of the welding process and make it difficult to achieve high-quality welds [3], [4]. One of the most critical issues is the formation of keyhole-induced porosity, an internal defect that compromises weld strength and structural integrity. Keyhole-induced porosity is one of the most critical defects in laser welding of aluminum alloys. It occurs when the vapor-filled keyhole, formed under high-energy laser irradiation, becomes unstable and

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collapses, trapping metal vapor or shielding gases within the molten pool. These trapped gases form pores as the weld solidifies, compromising mechanical integrity. Fluctuations in laser power density, melt pool dynamics, surface tension gradients, and recoil pressure imbalances often trigger keyhole instability[5], [6], [7]. Recently, scholars have aimed to control or predict pore formation by analyzing and optimizing keyhole behavior. Several studies have focused on how process parameters influence keyhole stability and, consequently, porosity. For instance, Nisar et al. [8] optimized laser parameters for A5083 aluminum alloy, demonstrating that suitable parameter combinations can significantly reduce keyhole-induced porosity. Similarly, Li et al.[9] explored multi-objective optimization of laser welding conditions for AA6061, highlighting trade-offs between welding quality and energy consumption. They found that it implicitly supports porosity control by stabilizing process parameters. Dhanaraj and Rathinasuriyan[10] improved weld quality in AA7075-T6 through fiber laser parameter adjustment, indirectly reducing the likelihood of pore formation. Khan et al.[11] demonstrated that laser beam defocusing alters the weld pool geometry and stabilizes the keyhole, effectively preventing concavity and gas entrapment. More recently, Saedi Ardahaei and Pham [12] performed a numerical study on keyhole shape and penetration depth under different power modulation schemes, emphasizing how pulse shaping can mitigate keyhole collapse and enhance weld uniformity. In addition, Meena et al. [13]conducted a combined numerical and experimental study of AA1050 laser welding, demonstrating how variations in laser power and feed rate affect keyhole depth, width, and associated porosity formation. Their findings indicate that insufficient energy density results in unstable keyholes and increased spatter, underscoring the importance of monitoring keyhole stability to prevent porosity and ensure weld quality. Saedi Ardahaei and Pham[14] conducted a comprehensive numerical study to identify the physical forces governing keyhole instability in aluminum laser welding. Their findings emphasized the roles of recoil pressure, Darcy's damping, and surface tension in keyhole collapse and porosity formation. They demonstrated that specific pulse shapes, such as triangular or ramped rectangular waves, could stabilize keyhole dynamics and reduce defect risk[6].

Table 4. Performance metric of model.

Despite significant advances in understanding and modeling keyhole behavior, most existing studies rely on offline measurements or numerical simulations that lack integration with real-time data from the welding process. These approaches are often constrained to post-process analysis, which fails to capture the transient, highly dynamic behavior of the melt pool and keyhole as welding progresses. Since porosity formation is strongly linked to momentary instabilities in the keyhole, the inability to monitor and analyze these phenomena in real-time limits the practical application of these insights in manufacturing environments. In recent years, the integration of artificial intelligence (AI) into laser welding manufacturing and monitoring systems has emerged as a promising direction, enabling more responsive and adaptive quality control frameworks [15], [16]. Some studies focused on the vision-based keyhole and molten pool monitoring, where high-speed imaging has emerged as a powerful tool to observe rapid keyhole and molten pool dynamics during welding. Wen et al. [17] employed side-view high-speed imaging through transparent quartz glass to measure keyhole depth and inclination angle in pulsed laser welding of 6061 aluminum. By extracting these features using image enhancement and Canny edge detection, they demonstrated that pulse frequency was the most influential parameter in determining keyhole stability and weld penetration. Mihai et al. [18] monitored plume characteristics using imaging at up to 30,000 fps, showing that plume area and grayscale intensity correlated with internal porosity. Zhang et al. [19] used a coaxial high-speed camera in combination with CNN-based classification to detect porosity in aluminum welds, achieving over 96% accuracy based on labeled cross-sectional pore data. Kim et al. [20] introduced a dual-view coaxial monitoring system for Al 5052, applying YOLOv4 and ResNet models to detect keyhole apertures and predict laser absorptance, revealing the impact of shielding gases like helium on stabilizing the weld. In a similar approach, Kim et al. [21] examined cold-rolled steel welding with synchronized top and bottom views to analyze absorptance drops during defect formation, achieving real-time feedback with a latency of 1.66s. Zhang et al. [22] addressed the challenge of metal vapor interference by introducing an adaptive fusion strategy that combined 10 frames into a clearer composite image. This enabled a CNN

classifier to achieve penetration state recognition with 99.86% accuracy, demonstrating robust real-time capability despite challenging optical conditions.

To overcome the limitations of vision-only systems, several studies have employed hybrid sensing strategies. Ma et al. [23], [24] combined high-speed imaging with coherent optical systems to simultaneously track keyhole surface and depth features. They applied wavelet packet decomposition (WPD) to derive spectrograms that reflected porosity-associated fluctuations, which were then classified using deep CNNs. These systems achieved detection accuracies above 90%, confirming that instability in keyhole geometry strongly predicts porosity. Shevchik et al. [25] integrated laser back-reflection and acoustic emission signals with deep learning, guided by X-ray radiography for ground truth. Their CNN models classified transient events, such as pore formation and keyhole blowout, with high accuracy and operated at millisecond-scale latency. Li et al. [26] proposed a hybrid monitoring system for Al-Cu lap welding using OCT and spectral diagnostics, training a BPNN to predict seam width with an error of just 10.7 μm , substantially outperforming single-sensor models. Zhang et al. [27] employed a complex sensor suite including visual, UV-visible, spectral, and photodiode inputs to track weld defects in stainless steel. Their SSAE-GA model effectively identified conditions such as blowouts and undercuts with over 95% accuracy across varied welding conditions. These studies collectively demonstrate that fusing diverse sensing modalities can enhance monitoring robustness and provide multidimensional insight into weld quality. With the rise of intelligent manufacturing, AI and deep learning models have become central to weld defect prediction. CNN architectures were applied extensively across studies to process either images or spectrograms derived from weld signals. Ma et al. [24] used CNNs to analyze time-frequency representations of keyhole depth and area signals, achieving 90.5% classification accuracy for porosity segments. Zhang et al. [19] trained a simplified AlexNet-style CNN on weld-pool images aligned with cross-sectional pore labels, reaching 96.1% accuracy despite challenges in detecting sub-150 μm pores. Ma et al. [28] developed a real-time porosity classification model using a Deep Belief Network (DBN) trained on denoised keyhole depth signals, achieving >98% accuracy and proposing

a process planning database for industrial deployment without retraining. Ma et al. [29] advanced the field by introducing a multi-fidelity deep learning framework combining simulation-derived and sensor-acquired data. Their tandem DBN architecture fused maximum temperature and keyhole depth features, achieving an RRMSE of 0.0574 and a latency under 20ms. Ma et al. [30] and Zhang et al. [27] optimized ANN and SSAE models using genetic algorithms, incorporating thermal history and feedback mechanisms to enhance model stability and convergence. These AI-driven approaches significantly improve monitoring precision and enable real-time decision-making, even under varying process conditions.

While many studies focus solely on monitoring, fewer have implemented active control strategies. A key contribution in this area is from Huang et al. [31], who integrated arc spectral sensing with a fuzzy-PID control system in GTAW of 5A06 aluminum. Their system employed Locally Linear Embedding (LLE), wavelet packet transform (WPT), and Empirical Mode Decomposition (EMD) to extract porosity indicators from spectral data. An ELM classifier achieved 96.2% accuracy, and the fuzzy-PID controller dynamically adjusted peak current based on hydrogen-to-argon intensity ratios, thereby reducing or eliminating porosity in real-time. Their validation experiments demonstrated robustness across a range of groove angles and thicknesses. Although most other systems (e.g., Ma et al. [28], Zhang et al. [22]) focused on passive monitoring, they laid the groundwork for future closed-loop control architectures. The integration of detection, classification, and actuation, as demonstrated by Huang et al. [31], marks a pivotal shift toward adaptive manufacturing systems capable of autonomous quality assurance.

Based on the critical analysis of the literature presented above, three overarching research gaps have been consistently identified. First, while several studies have explored individual aspects of laser welding quality, such as keyhole geometry, melt pool dynamics, or defect classification, few have proposed a unified, real-time monitoring system that integrates image acquisition, defect labeling, feature extraction, and machine learning within a closed-loop framework. Second, although porosity detection methods have advanced significantly, most rely on post-process or destructive validation,

highlighting a lack of intelligent, non-destructive evaluation approaches that can be deployed in real-time. Third, despite the widespread use of high-speed imaging for visualizing keyhole behavior, the automated extraction of geometrical features relevant to porosity prediction remains underdeveloped, with most studies still depending on manual analysis. These three limitations, drawn directly from the body of literature discussed, reflect a persistent lack of system-level integration, real-time capability, and automation in current welding quality monitoring systems.

- **Absence of Integrated Real-Time Monitoring Frameworks:** Despite progress in machine learning–based monitoring and imaging, an integrated, real-time system for porosity detection, combining high-speed imaging, defect labeling, feature extraction, and physical validation, remains largely unexplored. Most existing studies focus on isolated components, such as classification or post-process analysis, with limited emphasis on closed-loop control or ground-truth validation. While recent studies, Xu et al. [32], have linked keyhole morphological instability to porosity, few have translated these insights into scalable, validated solutions for industrial implementation. A framework capable of predicting weld quality in real time and adjusting process parameters adaptively would offer significant industrial value. The method proposed in this study represents an essential step in this direction.
- **Lack of Intelligent Non-Destructive Evaluation in Keyhole Porosity Analysis:** Studies exploring keyhole-induced porosity in laser welding and additive manufacturing often rely on destructive cross-sectional analysis to validate correlations between keyhole dynamics and pore formation [33], [34]. Although in situ synchrotron X-ray imaging has advanced understanding of keyhole fluctuation, collapse, and bubble evolution [34], [35], such methods remain resource-intensive and unsuitable for in-process monitoring. This study introduces a novel approach that extracts spatial and geometric features of the keyhole to enhance physical understanding of

keyhole behavior, without relying on destructive validation, to assess porosity risk through keyhole stability. This real-time capability offers a practical and scalable solution for enhancing weld quality control.

- **Automatic Geometrical Features Extraction of Keyhole:** Although high-speed imaging is widely used to visualize keyhole behavior in laser welding, there is a lack of automated methods for extracting geometrical features relevant to porosity prediction. Most studies rely on fully manual analysis or post-process inspection, limiting real-time application. Few have quantified keyhole metrics, such as aspect ratio or centroid shift, which correlate with defect formation. This study addresses this gap by introducing a semi-automated pipeline that extracts keyhole features from high-speed thermal images, enabling machine learning–based, in-process porosity detection.

In the specific context of aluminum laser welding monitoring, several studies have proposed diverse sensing and analytical approaches for porosity detection, yet each presents distinct limitations regarding real-time applicability, physical validation, or system complexity. For instance, Katayama et al. [36] and Kawahito et al. [37] employed dual-signal acquisition systems, combining reflected light and thermal radiation—with high-speed imaging to identify surface defects such as underfill and spatter. However, their methods lacked real-time porosity detection and were constrained by limited validation against internal defects. Turichin et al. [38] introduced a multisensory monitoring framework integrating photodiodes and CCD imaging for defect detection, but the system relied on post-process X-ray analysis and presented challenges for industrial scalability. Kim et al. [20] and Nam et al. [39] utilized coaxial imaging and deep learning models (YOLOv4, ResNet, YOLOR-CSP) to monitor keyhole absorptance and geometry, achieving high predictive accuracy, though their methods required simulation-based data and were not validated with actual porosity measurements. Sibillano et al. [40] and Wen et al. [17] explored optical and keyhole morphology indicators for tracking welding mode transitions and penetration depth, without extending their frameworks to porosity

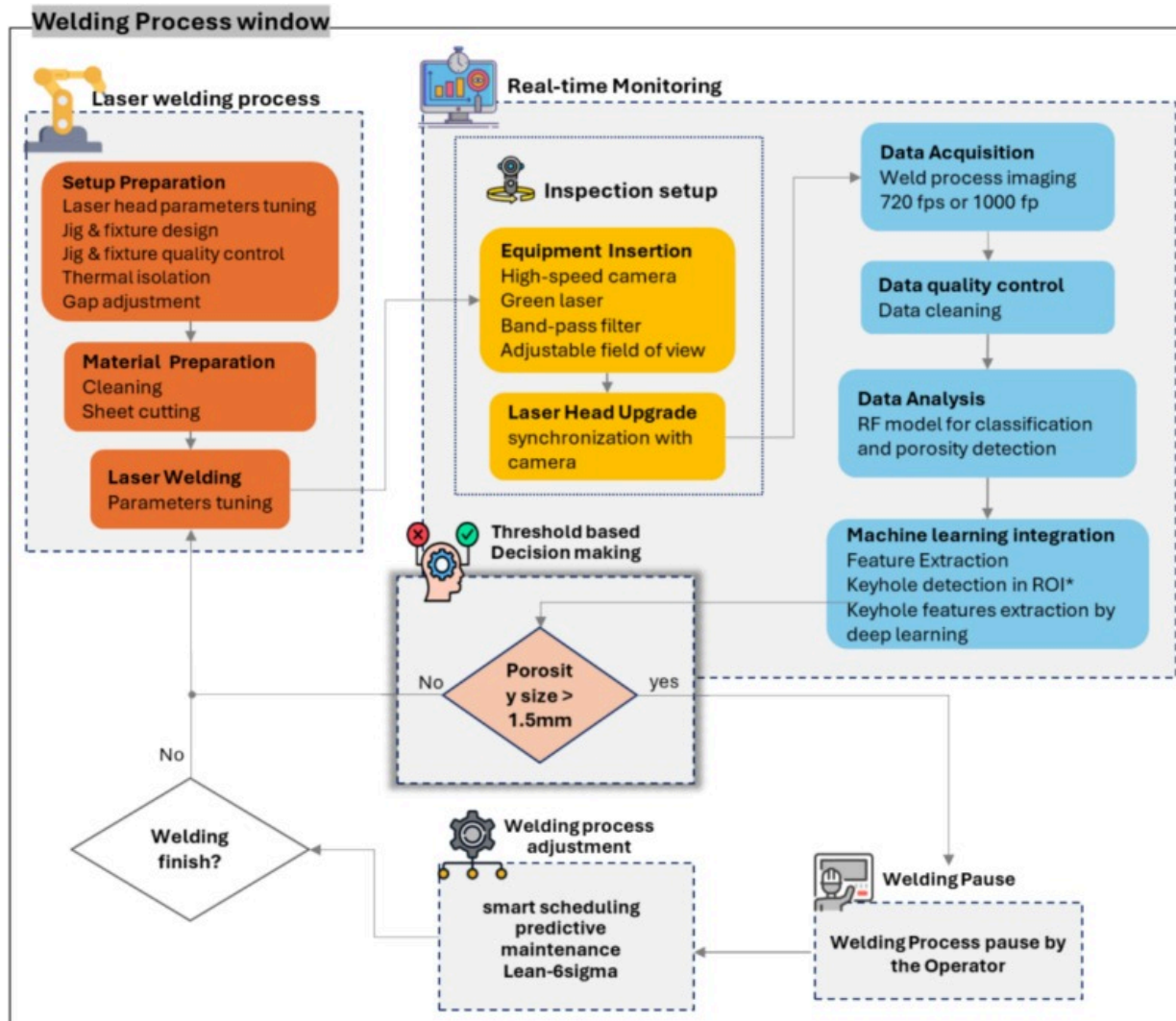
monitoring. Ma et al. [30] developed a sophisticated real-time porosity detection system based on 3D keyhole morphology, ensemble empirical mode decomposition, and ANN-GA optimization, yet the model's temporal dependence and high computational demands limit its real-time deployment potential. In contrast, the present study introduces a streamlined, interpretable framework using coaxial high-speed imaging, semi-automated geometric feature extraction, and Random Forest classification, physically validated through synchronized X-ray inspection. This approach enables non-destructive, frame-level porosity prediction without reliance on complex simulation inputs or multi-sensor fusion, offering a more practical and scalable solution for in-process quality assurance in aluminum laser welding.

While recent advancements have significantly improved the understanding of keyhole dynamics and porosity formation in laser welding, most existing studies remain limited to post-process inspection, numerical simulations, or vision-based monitoring without full real-time integration. Many approaches rely on destructive validation methods or offline modeling, which are not suitable for adaptive control in industrial settings. Although AI and Machine/Deep learning (ML/DL) techniques have shown high accuracy in defect classification, they are often applied in isolated scenarios, without physical validation, real-time processing, or practical scalability. Furthermore, few studies leverage synchronized top-view imaging of the melt pool combined with ML and ground-truth X-ray data to detect porosity as it forms. Therefore, there is a need for an integrated, real-time monitoring framework that enables direct, non-destructive assessment of porosity during welding and establishes the foundation for closed-loop control of the laser welding system and its parameters.

The objective of this study is to develop an ML-based framework for predicting keyhole-induced porosity formation in laser-welded aluminum joints. The model targets weld segments with porosity exceeding a critical threshold of 1.5mm, which corresponds to the maximum permissible limit defined for quality level D in ISO 13919-1:2019(E) [41]. Prediction is performed by analyzing high-speed visual data of the weld line, from which geometric features of the keyhole are extracted. These features serve as inputs to a

Random Forest classifier trained to distinguish between porous and non-porous weld lines. By emphasizing real-time image acquisition and feature-based classification, this work contributes to the foundation for intelligent, non-destructive quality monitoring in laser welding applications.

Fig. 1 illustrates the conceptual framework of the proposed intelligent porosity monitoring system, focusing on the welding process window. This includes three key stages: setup preparation, real-time inspection, and threshold-based decision-making. During welding, a high-speed camera system, integrated into the laser head, captures top-view images of the weld pool. These images undergo a series of sequential processing steps, including data acquisition, data cleaning, feature extraction and measurement, and porosity classification using a Random Forest model. If the predicted porosity exceeds the 1.5mm threshold, the system flags the weld segment as defective. While the present study is limited to the development and validation of this detection framework, **Fig. 1** also outlines a prospective extension toward closed-loop control. In this envisioned setup, welding would continue automatically when no porosity is detected and pause for process adjustment when porosity is identified. This forward-looking component is included to demonstrate how the proposed framework serves as a foundational step toward real-time control and continuous quality assurance in industrial laser welding applications.



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Fig. 1. Flowchart of the proposed real-time porosity monitoring system using high-speed imaging and machine learning. (* Region of interest (ROI)).

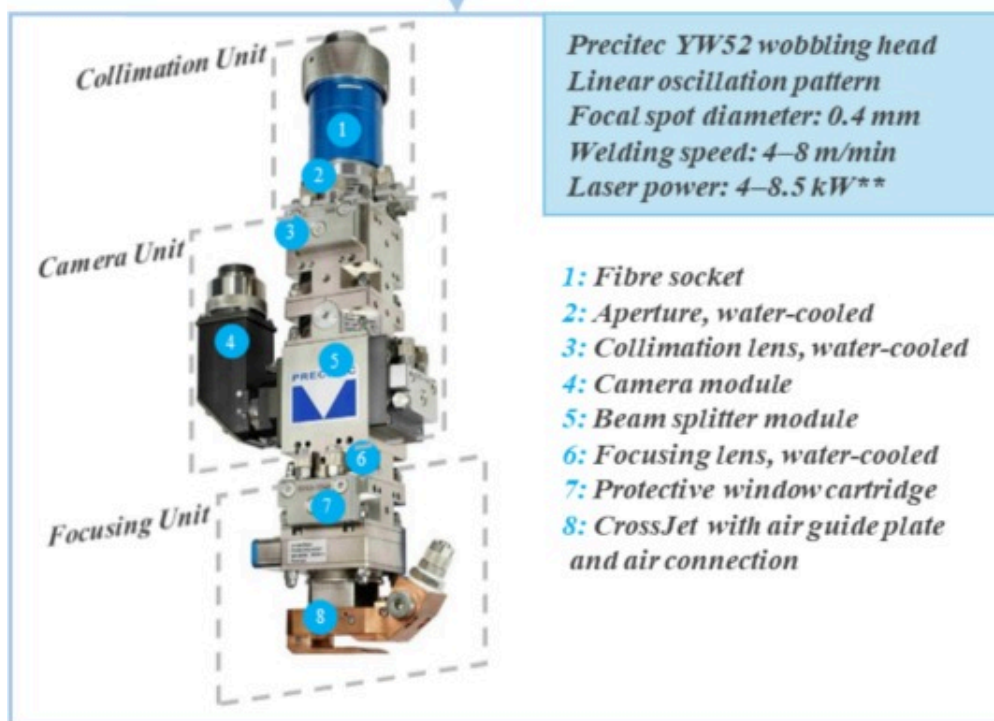
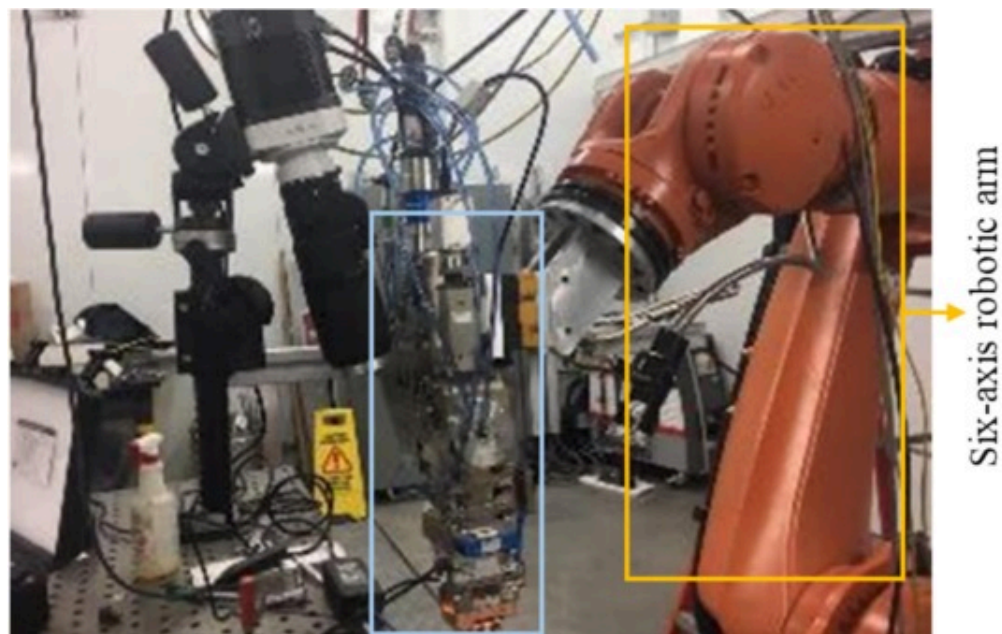
2. Materials and methods

2.1. Laser welding setups

In this study, AA6061-T6 aluminum alloys were used in an overlap configuration, with thicknesses ranging from 1.6mm to 2mm. The choice of AA6061-T6 is based on its precipitation-hardened composition, primarily consisting of magnesium and silicon, which provides superior weldability, corrosion resistance, and an optimal strength-to-weight ratio. These characteristics make it widely used in various industrial applications. The chemical composition and mechanical properties of the alloy are summarized in [Table 1](#). Single lap joint (SLJ) laser welding was performed using a Trumpf TruDisk 10kW solid-state laser, equipped with a Precitec YW52 wobbling head and a SCANLAB intelliWELD 30 FC scanner ([Fig. 2](#)). The laser beam, transmitted through a 200- μ m fiber, was focused to a 0.4mm spot size. An overlap configuration with zero interface gap between plates was used to intentionally promote porosity formation, enabling the development of a robust porosity monitoring model by capturing a wide range of defect conditions. Welding was conducted at a fixed speed of 50m/min and a laser power of 5.5kW, based on literature guidance and equipment validation, ensuring optimal surface quality. These parameters, detailed in [Table 2](#), were selected during preliminary experiments and kept constant throughout the study.

Table 1. Chemical composition of AA 6061-T6 (% wt) [\[42\]](#).

Chemical composition	Al	Cr	Cu	Fe	Mg	Mn	Other, total	Si	Ti	Zn
% wt	0.15–0.35	0.1	0.4	2.2–2.8	0.1	0.25	0.1	0.4–0.8	Max 0.15	Max 0.25



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Fig. 2. Laser welding machine and monitoring setup.

Table 2. Laser welding process parameters of YW52 welding head.

ID	Laser power (kW)	Travel speed (m/min)	Oscillation amplitude (mm)	Oscillation frequency (Hz)
#1	5.5	5.0	0.5	200

2.2. Monitoring setups and method

The monitoring configuration is presented in [Fig. 2](#). A MIRO M310 high-speed camera was employed to record the welding process at a frame rate of 1000 frames per second and a spatial resolution of 640×480 pixels. The camera was coaxially integrated with the laser head and synchronized with the welding control system, ensuring accurate temporal correspondence between the visual observations and the process data.

The 532nm narrow band-pass optical filter was mounted coaxially on the MIRO M310 camera lens, directly aligned with the optical axis of the laser head to ensure spatial coincidence between the observed keyhole region and the laser interaction zone. The filter allowed transmission of the auxiliary 532nm illumination while suppressing broadband thermal radiation from the molten pool. Synchronization between the galvanometer scanner, the TruDisk laser source, and the high-speed camera was achieved through a common TTL-trigger signal distributed by the Precitec YW52 controller. Each trigger pulse initiated both frame acquisition and galvanometer motion commands, yielding a temporal synchronization accuracy better than 0.5ms. Spatial calibration was performed using etched fiducial marks to maintain alignment within $\pm 25\mu\text{m}$ across the 6mm field of view. In this study, the oscillation amplitude (0.5mm) and trajectory were kept constant to focus on feature–porosity correlation; future work will systematically

vary these parameters to examine dynamic keyhole responses and their quantitative linkage to pore formation.

A single weld line was produced specifically for data collection. During welding, the high-speed camera continuously recorded the top-view dynamics of the keyhole and melt pool. To enable localized analysis of keyhole dynamics, the weld line was divided into 6mm segments. This segmentation scale is motivated by prior high-speed imaging studies, which consistently analyze keyhole instabilities and porosity mechanisms, such as keyhole collapse, vapor jet oscillations, and spatter ejection, within fields of view ranging from 2.5mm to 3.3mm. These mm-scale windows have proven sufficient for capturing the spatial and temporal resolution required to observe defect-inducing fluctuations. Accordingly, a 6mm segment length balances localized diagnostic resolution with practical coverage, ensuring both analytical precision and compatibility with industrial weld evaluation frameworks [30], [33], [34], [35]. Post-weld radiographic inspection, conducted in accordance with ISO 13919–2 standards, served as ground truth for porosity characterization. Two 6mm segments were selected: one segment containing porosity ≥ 1.5 mm in diameter (labeled as “Fail”) and one porosity-free segment (labeled as “Pass”). Each 6mm section corresponded to 720 consecutive frames. Besides, image processing and analysis were performed using ImageJ along with custom Python and R scripts. In this step, automated geometrical feature measurement using image-processing-based techniques are used. However, regarding the complexity of the welding pool and keyhole geometrical features, human-annotation-based techniques are required and used to improve the measurement accuracy. A total of 22 geometric and morphological features were extracted from each frame using the Region of Interest (ROI) Manager. These included parameters (presented in Table 3), such as area, perimeter, centroid position, circularity, aspect ratio, solidity, and Feret diameters. Additionally, optical flow estimation was applied to track pixel-level motion across frames, allowing for the assessment of transient instabilities. The extracted features were used to train a Random Forest (RF) classifier, selected for its robustness, interpretability, and ability to handle high-dimensional data without extensive preprocessing. The model was evaluated

using a standard 80/20 cross validation test to assess generalization performance. Importantly, the RF algorithm provides feature importance rankings, which help identify the melt pool characteristics most strongly associated with porosity formation. This classification model serves as the basis for the proposed in-process porosity detection system that eventually contributes to the development of a real-time porosity monitoring framework integrating high-speed imaging, spatial feature extraction, and ML to detect porosity during the laser welding of aluminum alloys. The system captures dynamic melt pool and keyhole behavior, correlating visual features with internal defects confirmed by X-ray imaging.

Table 3. Extracted feature from ROI.

Feature name	Description
Area	Size in square pixels
X, Y	Centroid coordinates
XM, YM	Center of mass (brightness-weighted)
Perimeter	Boundary length
BX, BY	Bounding rectangle coordinates
Width, Height	Bounding box dimensions
Major, Minor	Ellipse fit axes
Angle	Ellipse orientation
Circularity	$4\pi \times \text{Area} / \text{Perimeter}^2$
Roundness	$4 \times \text{Area} / (\pi \times \text{Major}^2)$
Aspect Ratio	Major/Minor axis
Solidity	Area/Convex area

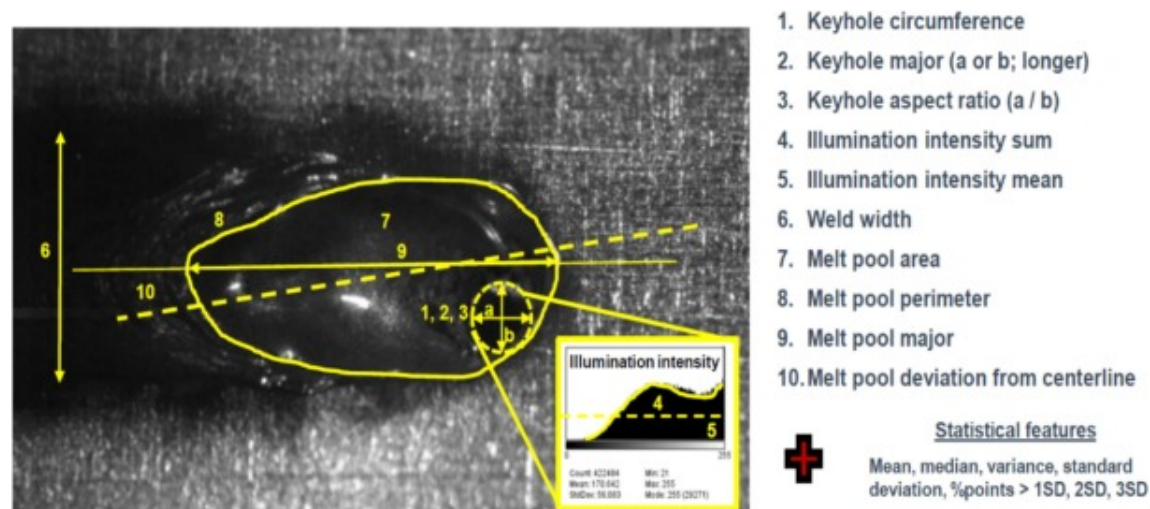
Feature name	Description
Feret	Max caliper distance
Feret Angle	Orientation of Feret diameter
FeretX, FeretY	Starting coordinates of Feret diameter
MinFeret	Minimum caliper distance

3. Results

3.1. Image-based monitoring of keyhole characterization

Fig. 3 shows a high-speed image captured after the welding process, focused on the solidified weld pool. A distinct keyhole-like structure is visible at the center of the weld track, representing a well-recognized feature in aluminum laser welding. This feature typically forms due to intense laser-material interaction, where rapid melting and metal evaporation lead to the development of a deep, narrow cavity. Recoil pressure from vaporization, combined with surface tension gradients, induces Marangoni convection within the molten pool. In aluminum alloys, Marangoni flow typically moves outward from the center, which can broaden the melt pool and destabilize the keyhole under steep thermal gradients [43]. When this flow becomes excessive, the keyhole may collapse or fluctuate irregularly, promoting gas entrapment and porosity formation. The residual shape of the keyhole, often appearing as a balling or semi-squared feature, remains visible along the weld surface. These patterns reflect the molten pool dynamics and can be used as visual indicators of internal instability. From this image, a set of geometric and intensity-based features was extracted using image analysing, in this study ImageJ software. These include the keyhole area, representing the footprint of the cavity; major axis length, reflecting elongation and orientation; and aspect ratio, which captures the shape and stability of the keyhole. Additionally, mean and total illumination intensity were computed to approximate energy absorption during welding. These parameters,

which together describe the spatial geometry and thermal signature of the keyhole, are essential for correlating surface characteristics with subsurface porosity. They form the basis for predictive modeling aimed at detecting weld defects based on visual and thermal cues.



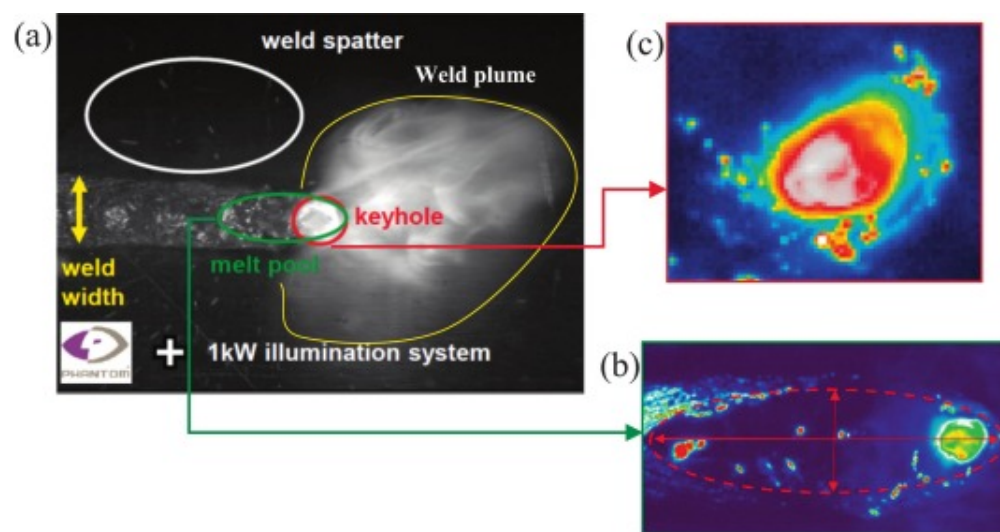
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Fig. 3. High-speed camera image taken after the welding process, showing the keyhole and melt pool features used for porosity analysis.

For real-time monitoring, the keyhole features remain visible during the welding process, as shown in Fig. 4a. This high-speed camera image captures the active weld zone, including several critical phenomena. The weld plume, generated by the interaction of vaporized metal and shielding gas, appears above the keyhole and indicates the intensity of vaporization. The weld spatter, which consists of ejected molten droplets, is also visible and reflects the dynamic instability of the molten pool. More importantly, the image highlights two core regions: the melt pool, where localized fusion occurs, and the keyhole, which is distinctly visible with high contrast at the center of the image. This clear visualization of the keyhole allows an assessment of its morphology and behavior.

To enhance clarity and improve feature detection, thermal filtering was applied to the same high-speed image. As shown in Fig. 4b and Fig. 4c, the thermal-filtered versions highlight the intensity distribution across the melt pool and provide a zoomed-in view of the keyhole region, respectively. These filtered images enable more precise analysis of thermal gradients, energy concentration, and the structural definition of the keyhole during the welding process.



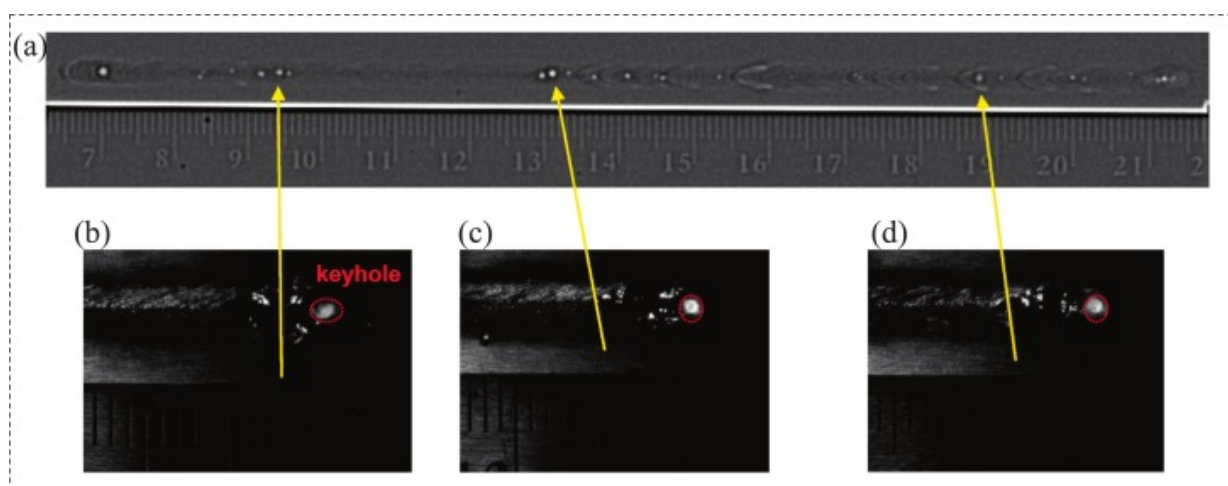
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Fig. 4. (a) High-speed camera image during welding showing the weld plume, spatter, melt pool, and keyhole. (b) Thermal-filtered image of the melt pool. (c) Zoomed thermal-filtered image of the keyhole region for enhanced feature clarity.

To experimentally validate the real-time detection of porosity-prone regions, X-ray radiography was performed after welding and compared with high-speed thermal images captured during the process. Fig. 5a shows the X-ray image of the entire weld seam, where internal pores are identified as white spots along the weld line. These porosity locations were then matched with the same regions observed during welding in the thermal images shown in Fig. 5b–d. These frames, captured in real-time, display the

keyhole region with high thermal contrast, particularly in locations where porosity later appeared in the X-ray. The purpose of this step was to validate the effectiveness of the high-speed imaging approach. The matching between the real-time keyhole signatures and the confirmed porosity locations demonstrates that thermal images can successfully indicate defect-prone areas during welding. This correspondence indicates that, when keyhole behavior is detected in real-time, post-process X-ray inspection may no longer be necessary for quality assurance. In this study, the X-ray analysis was used only for validation purposes, confirming that keyhole instability detected during welding is a reliable indicator of internal porosity.



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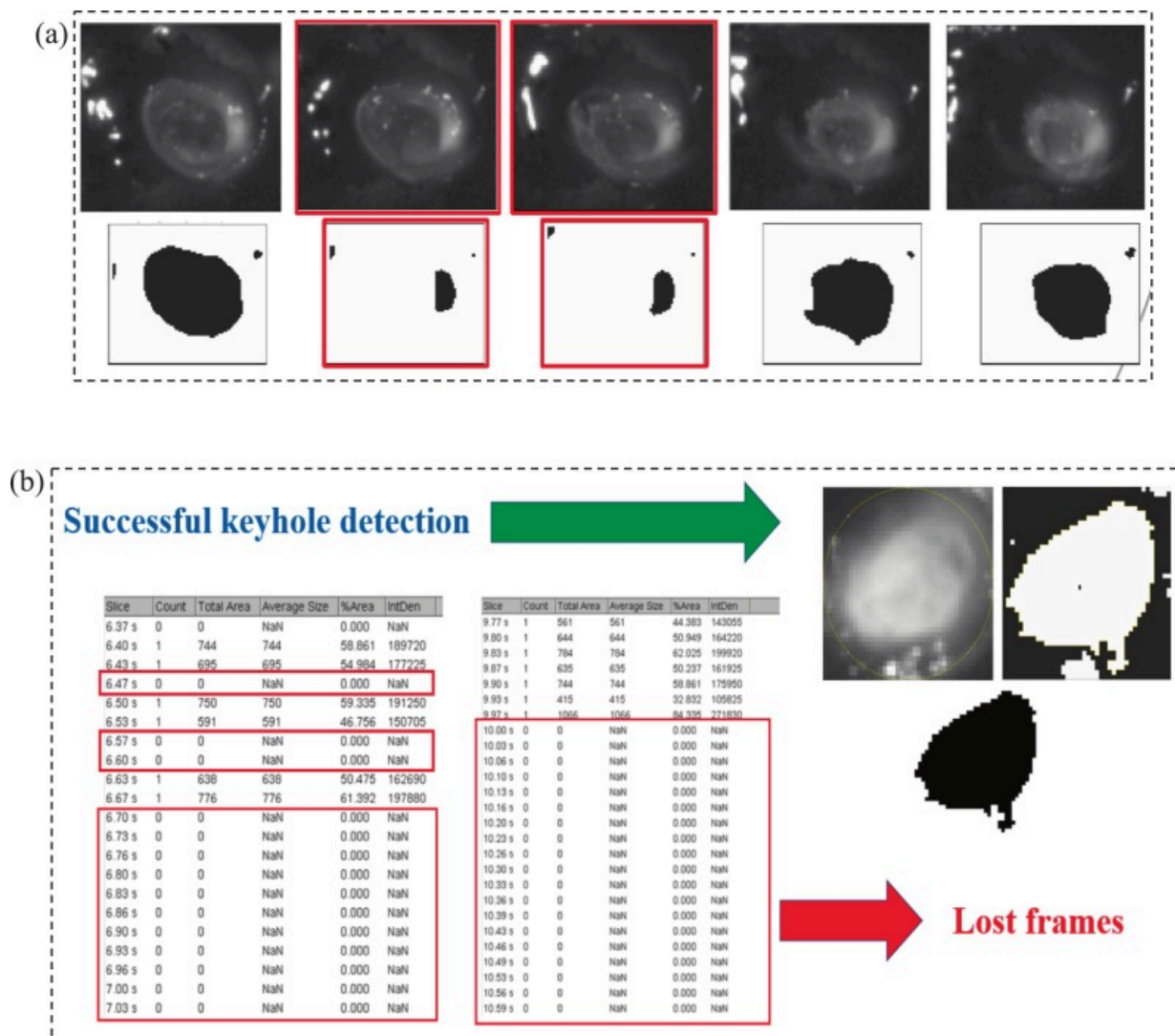
Fig. 5. (a) X-ray image showing porosity along the weld line. (b–d) Real-time thermal images of corresponding regions with keyhole instability linked to porosity.

Following the validation of porosity-prone zones through X-ray inspection, a detailed frame-by-frame image analysis was conducted to extract quantitative keyhole features from high-speed video data. For each 6mm weld segment, 720 individual frames were processed using ImageJ to capture temporal evolution and fluctuation of the keyhole. The extracted parameters serve as the primary descriptors of keyhole geometry and thermal

response as previously defined in Fig. 3. These metrics form the core feature set used in downstream classification and time-series modeling for porosity prediction.

Primary, the automated feature extraction process was implemented using a combination of threshold-based binarization, contrast enhancement, and morphological filtering techniques within ImageJ. This enables the segmentation of keyhole regions based on their thermal and geometric contrast relative to the surrounding weld pool. As shown in Fig. 6a, approximately 85% of the frames yielded successful segmentations, producing binary masks with well-defined contours suitable for reliable feature computation. These frames consistently delivered measurable and stable output values for area, shape, and intensity. However, segmentation failures occurred in the remaining ~15% of frames, outlined in red in Fig. 6a. As an initial step, the corresponding Fig. 6b presents ImageJ-generated data tables that summarize extracted features across the full frame sequence. Successfully segmented frames are characterized by measurable values for area and intensity, while lost frames are flagged with zero or NaN entries, indicating invalid data. These discontinuities affect the consistency of the feature dataset and introduce uncertainty in dynamic descriptors, such as temporal variance, fluctuation frequency, and keyhole deformation, all of which are critical for defect prediction. To this end, to improve segmentation success, preprocessing steps such as contrast normalization and Gaussian smoothing were applied to enhance feature boundaries and reduce high-frequency noise. These measures proved effective in moderately distorted frames but could not resolve severe degradation cases. Consequently, a human annotation step was implemented to recover feature data in lost frames. It begins with the collection of images requiring annotation, followed by the definition of clear annotation guidelines to ensure consistency. An appropriate annotation tool (e.g., CVAT, Labelbox) is then selected, and human annotators perform labeling tasks such as drawing bounding boxes, tagging objects, and reviewing the results. The labeled data undergoes quality control through peer review or AI-assisted checks. If errors are identified, the annotations are sent back for correction; otherwise, they are approved. The accepted annotations are then exported into standard formats (e.g., JSON, XML) for integration into the dataset. Using this

method, approximately 10–15% of the undetected frames were successfully restored, improving the overall dataset quality for subsequent training and evaluation.



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Fig. 6. (a) High-speed frames with detected and lost keyhole regions (in red). (b) ImageJ output showing segmentation results, with lost frames marked by zero or NaN values.

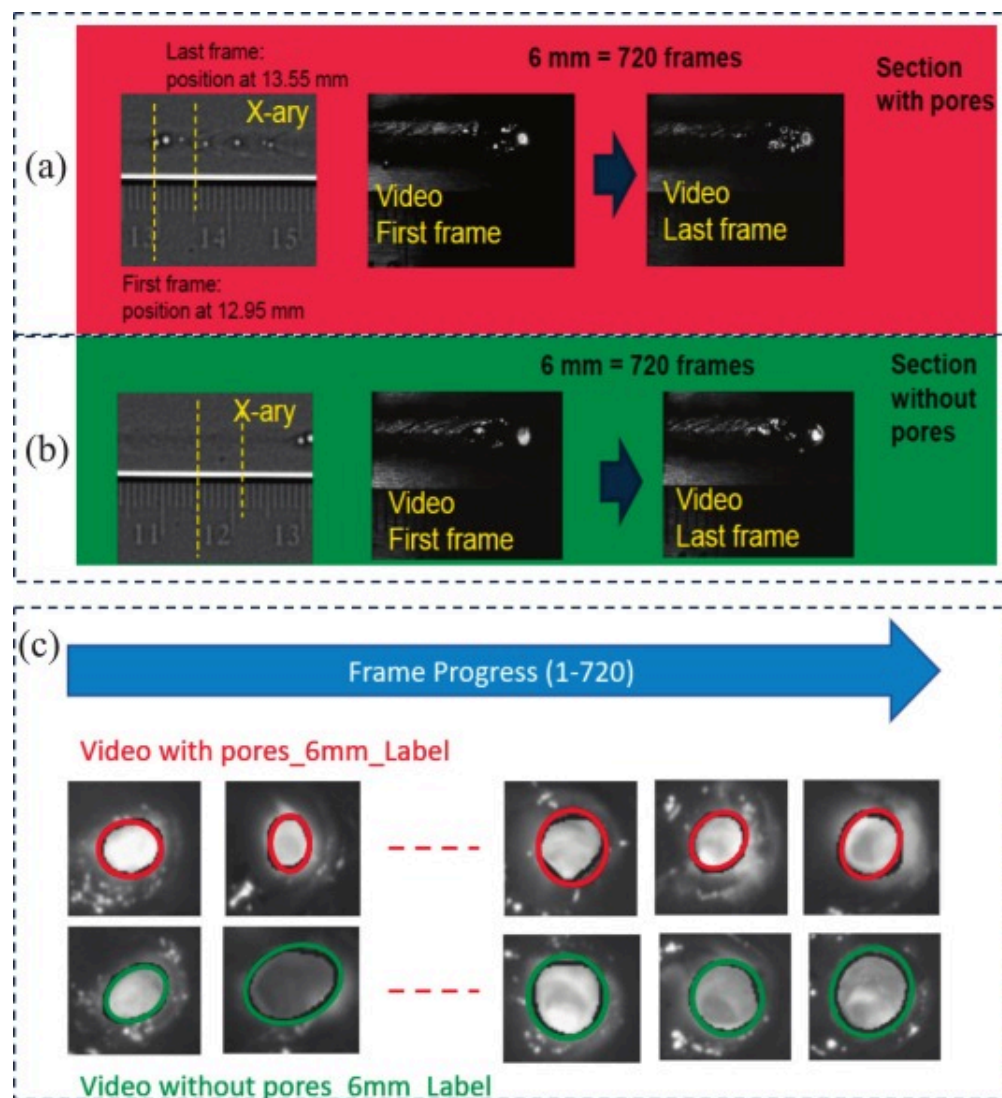
3.2. Frame by frame analysis

Due to the limitations observed in the automatic keyhole segmentation process, where approximately 15% of frames failed to yield valid binary masks ([Fig. 6](#)), a detailed frame-by-frame review was necessary to ensure dataset reliability. Although the automated extraction method successfully localized the keyhole in most cases, it was insufficient for directly inferring porosity formation. This is because porosity does not always result from visibly unstable melt pool behavior. The weld pool is a highly dynamic environment, and porosity arises from threshold-driven mechanisms such as vapor pressure accumulation, insufficient melt expulsion, or Marangoni flow stabilization. Visual indicators like keyhole elongation, asymmetry, or balling may occur without leading to actual porosity formation. Therefore, expert annotation was required to generate a reliable ground truth dataset for training and evaluating classification models.

Building on prior studies that have demonstrated a strong correlation between melt pool dynamics and porosity formation [[44](#)], [[45](#)], [[46](#)], [[35](#)], the current work employs coaxial high-speed imaging to extract relevant keyhole features and correlate them with internal porosity via X-ray inspection. Illumination contrast from these images was used to identify keyhole features, which were then spatially aligned with post-weld X-ray scans to verify whether they matched porosity regions. Frames showing elevated contrast and geometric disturbance overlapping with X-ray-verified porosity were labeled as “fail,” while those in porosity-free zones were labeled as “pass.”

This classification was performed on two 6mm weld segments (one defective and one defect-free) resulting in a total of 1,440 annotated frames. This enabled identification of keyhole dynamics most associated with porosity and formed the basis for the supervised learning model. A detailed human-based annotation strategy was developed to systematically label frames based on their correlation with verified porosity. As shown in [Fig. 7](#), the upper segment in [Fig. 7a](#) (red background) contains visible porosity between 12.99mm and 13.55mm, as confirmed by X-ray inspection. This image includes both the post-weld X-ray scan (top) and the corresponding top-view coaxial image of the weld

seam (bottom), aligned spatially to show the direct correlation between internal porosity and surface keyhole features. In contrast, the lower segment in Fig. 7b (green background) is porosity-free and represents stable welding conditions.



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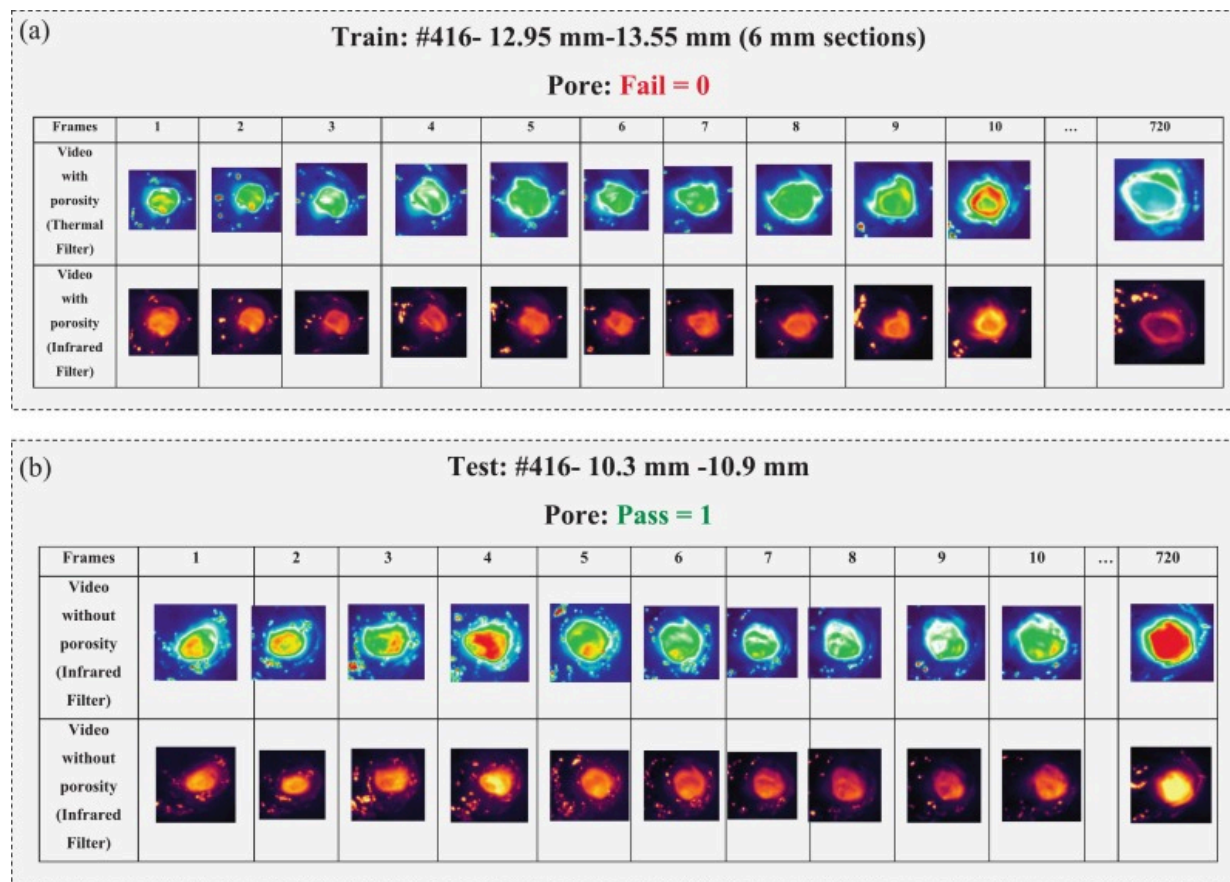
Fig. 7. (a) X-ray image showing porosity in the weld segment marked as “fail” (red); (b) X-ray image of defect-free segment marked as “pass” (green); (c) Annotated high-speed video frames for both segments, illustrating keyhole contours used for classification.

Each high-speed frame was aligned with its corresponding weld segment using spatial and temporal synchronization. The resulting annotated contours, enriched with contrast-based features, provided a reliable visual distinction between defective and non-defective welds. Representative examples of pass and fail keyhole frames are shown in Fig. 7c, demonstrating clear contrast and geometric differences that correlate with porosity occurrence.

This annotated dataset provides a robust foundation for supervised machine learning by linking real-time keyhole behavior with confirmed internal weld defects. The strong empirical correlation between coaxial thermal imaging and X-ray-verified porosity reinforces the credibility of the labels and enhances the generalizability of the trained models. Ultimately, this approach enables real-time porosity risk prediction based on visual features alone, eliminating the need for post-process X-ray inspection.

Fig. 8 illustrates selected frames from two annotated video sequences, one corresponding to a porosity-containing region (Fail=0), and the other to a defect-free region (Pass=1) of weld line #416. Each sequence consists of 720 frames extracted from a 6mm weld segment, which has been previously verified using X-ray inspection. To enhance the visual representation of keyhole behavior across the entire 6 mm weld segment, thermal and infrared filters were applied to the high-speed video frames. These filters were not used for analysis or model input but served only to improve the visibility of keyhole contrast and energy distribution during welding. In the failure region (Fig. 8a), the thermally and infrared-filtered frames reveal pronounced frame-to-frame fluctuations in keyhole geometry, brightness distribution, and boundary symmetry. These variations are indicative of instability in melt pool dynamics, such as uneven energy absorption, transient keyhole collapse, or excessive vapor recoil pressure. The irregularities are characterized by abrupt shifts in keyhole position, distorted outlines, and fluctuating

intensity levels, features commonly associated with conditions favorable for porosity formation. Such instabilities suggest the presence of localized thermal gradients and inconsistent fluid flow, which may trap gas or disrupt solidification in ways that result in voids. By contrast, in the pass region (Fig. 8b), the keyhole appears highly stable throughout the sequence. The frames show uniform thermal intensity, consistent geometry, and symmetrically distributed contours. Also, no abrupt deformations or spatial shifts are observed, indicating a balanced energy input and steady melt pool behavior. These visual signatures correlate strongly with the absence of porosity confirmed in the corresponding X-ray scan, supporting the reliability of visual thermal indicators for defect-free welding conditions.



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Fig. 8. (a) Porosity-containing segment (Fail=0) showing unstable keyhole behavior. (b) Defect-free segment (Pass=1) with stable geometry and intensity. Filters applied for visual clarity.

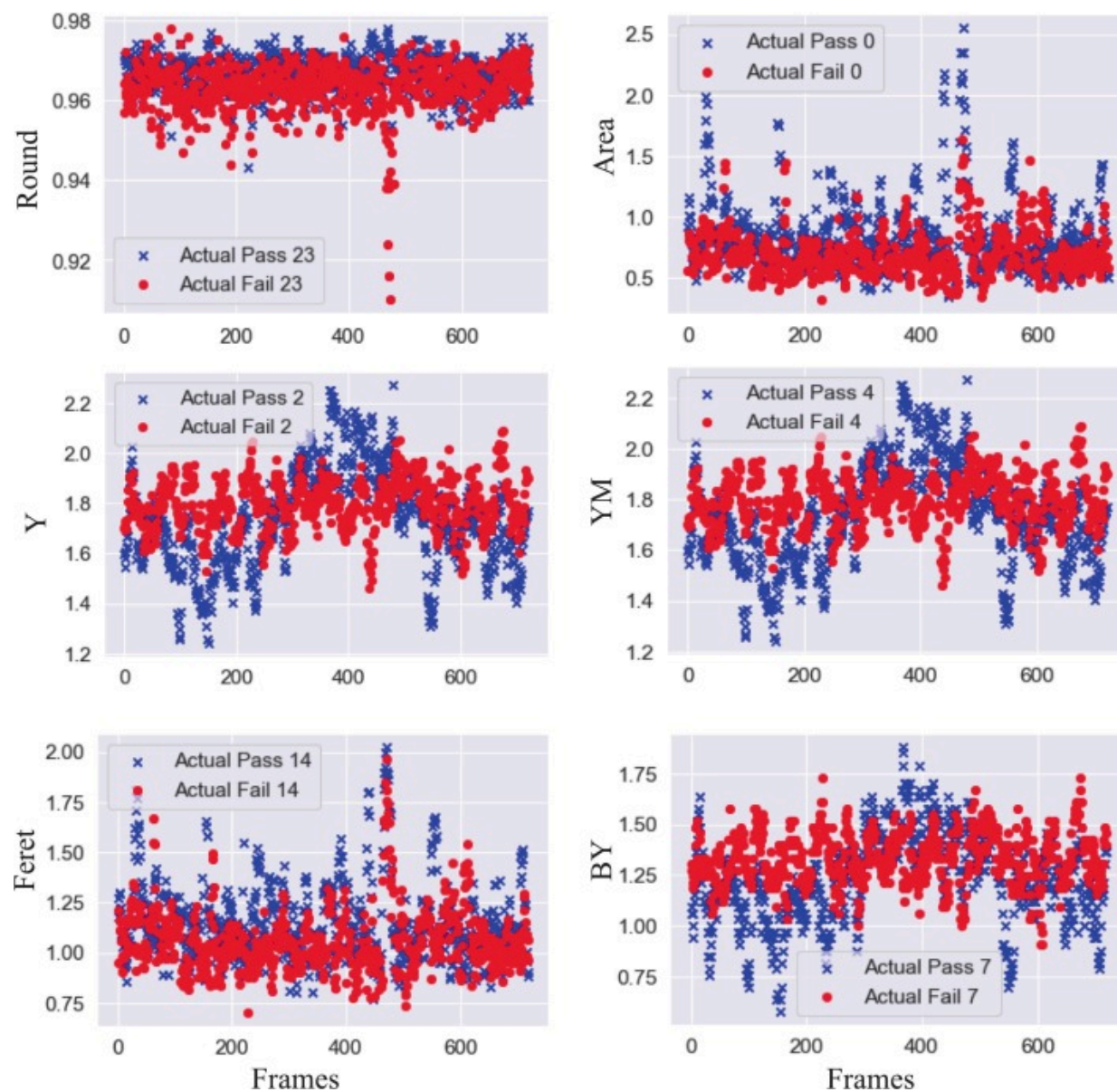
3.3. Feature extraction for machine learning input

Following the annotation of pass and fail sequences, a systematic feature extraction process was conducted to quantify the geometric and spatial properties of the keyhole for each of the 720 frames in both weld segments. The objective was to create a robust dataset of numerical features corresponding to keyhole behavior that could be used as input for machine learning classification models. To this end, ImageJ's ROI (Region of Interest) Manager was employed to define and analyze the keyhole area in each frame. The ROI represents the active region surrounding the laser-induced keyhole to determine weld quality. The ROI Manager enabled consistent tracking and measurement of keyhole properties over time by allowing the extraction of polygonal contours around the keyhole area and facilitating batch data processing. This image processing technique helped define and isolate regions of interest within each frame, making it possible to automatically analyze keyhole geometry and temporal changes across a series of images. A set of geometric and spatial features presented in [Table 3](#) was calculated for each frame to capture both the shape and energy distribution characteristics of the keyhole. The results of the following features are given in [Fig. 9](#):

- **Area:** Represents the total number of pixels within the detected keyhole boundary. It serves as a proxy for the molten pool size and the amount of absorbed energy. A sudden expansion or contraction in area may indicate keyhole instability or collapse, both of which are associated with pore formation.
- **Y (Centroid Y-coordinate):** Denotes the vertical position of the keyhole's geometric center. Variations in this value reflect keyhole motion or

oscillation in the vertical direction. A higher degree of fluctuation may signal unstable laser-material interaction and localized collapse.

- **YM (Center of Mass Y-coordinate):** Indicates the vertical center of intensity distribution within the keyhole. Since brightness correlates with thermal energy, YM provides insights into the energy concentration and its distribution within the melt pool. Shifts in YM may suggest asymmetrical energy input, leading to unstable vapor flow and porosity generation.
- **BY (Bounding Rectangle Y-coordinate):** Marks the vertical position of the top-left corner of the smallest bounding rectangle enclosing the keyhole. This feature tracks the upward or downward shift of the keyhole's entire geometry within the region of interest (ROI), and it can be sensitive to small vertical displacements in melt-pool positioning.
- **Feret's Diameter:** Defined as the longest distance between any two points along the keyhole boundary, this feature reflects the degree of elongation or stretching. A longer Feret's diameter may indicate keyhole asymmetry or collapse pathways, both of which can be linked to vapor instability and pore entrapment.
- **Roundness:** A dimensionless measure of how closely the keyhole shape approximates a perfect circle. Lower roundness values indicate elongated, irregular keyholes, which are often a sign of uneven energy input or recoil pressure fluctuations—both of which are potential precursors to porosity formation.



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Fig. 9. Extracted geometric features from annotated keyhole regions in Image J.

These extracted features were calculated using micro-coding within ImageJ's measurement functions, allowing for automated analysis of each ROI. The complete dataset, consisting of 720 frames per 6mm weld segment, each with quantified values for the six selected features, was visualized and analyzed to assess trends in keyhole behavior. Then, the mean values for each feature across the weld sequence were computed and are presented in Fig. 9. Although the dataset was primarily constructed using post-weld X-ray validation, further confirmation was obtained through metallographic cross-sections of a subset of welds. These supplementary evaluations confirmed the presence or absence of internal pores and were used to validate the consistency of image-derived features across multiple weld conditions. Statistical comparisons revealed that porous weld segments exhibited significantly higher temporal fluctuation in coaxial image intensity and greater variability in keyhole shape metrics. Specifically, Solidity, Feret's diameter, and Area were the most sensitive descriptors for distinguishing between sound and defective welds, indicating their relevance to melt pool instability. While internal porosity is not directly visible in coaxial surface images, it results from transient physical mechanisms, such as keyhole collapse, vapor entrapment, and turbulent melt flow—that simultaneously disturb the melt pool surface and thermal emission captured by the coaxial camera. These instabilities manifest as measurable fluctuations in image features, including brightness distribution, centroid position (Y, YM), elongation (Feret's diameter), and contour compactness (Solidity). Comparison with X-ray-verified weld segments confirmed that such fluctuations consistently appear in regions where pores ≥ 1.5 mm are present. Therefore, the extracted features encode surface manifestations of keyhole instability, reinforcing their physical relevance and supporting the model's ability to generalize beyond purely statistical associations. This strengthens the interpretability and applicability of the proposed monitoring framework for real-world welding conditions.

3.4. Machine learning implementation- Random Forest (RF) classification

A Random Forest (RF) classifier was implemented to evaluate the capability of machine learning in predicting porosity based solely on features extracted from high-speed

thermal imaging of laser welding. The model was developed using 1440 labeled Video frames as images, equally split between Pass (defect-free), and Fail (porosity-present) conditions, as determined through X-ray validation. Each sample represented features aggregated from 720 frames corresponding to a 6mm weld segment. The RF model was trained using Scikit-learn with the following configuration:

- **N_estimators=100:** A total of 100 decision trees were used to form the ensemble. This number strikes a balance between model performance and computational efficiency. Increasing the number beyond this point provided minimal gains during validation.
- **Criterion = 'Gini':** The Gini impurity metric was selected to evaluate split quality. It is the default and widely used criterion in classification problems due to its lower computational cost compared to entropy while delivering comparable results.
- **Max_features=5:** Limiting the number of features considered at each split helps reduce overfitting and introduces diversity among trees. This value was selected based on grid search optimization and provided improved generalization on the validation set.
- **Random_state=0:** A fixed random seed ensures the reproducibility of the results and consistent model behavior across training sessions.

The model used out-of-bag (OOB) error estimation for internal validation and was further evaluated on a held-out 20% test set. The classifier achieved a prediction accuracy of approximately 80%, demonstrating reliable classification performance between defect and non-defect categories. To identify which input features contributed most to the model's decisions, Permutation Feature Importance (PFI) was applied. This method calculates the drop in model performance when feature values are randomly permuted.

Features that cause a larger decrease in accuracy are considered more critical for prediction. The PFI analysis indicated that the most significant features included:

- Keyhole aspect ratio,
- Mean and total thermal intensity,
- Melt pool deviation from centerline.

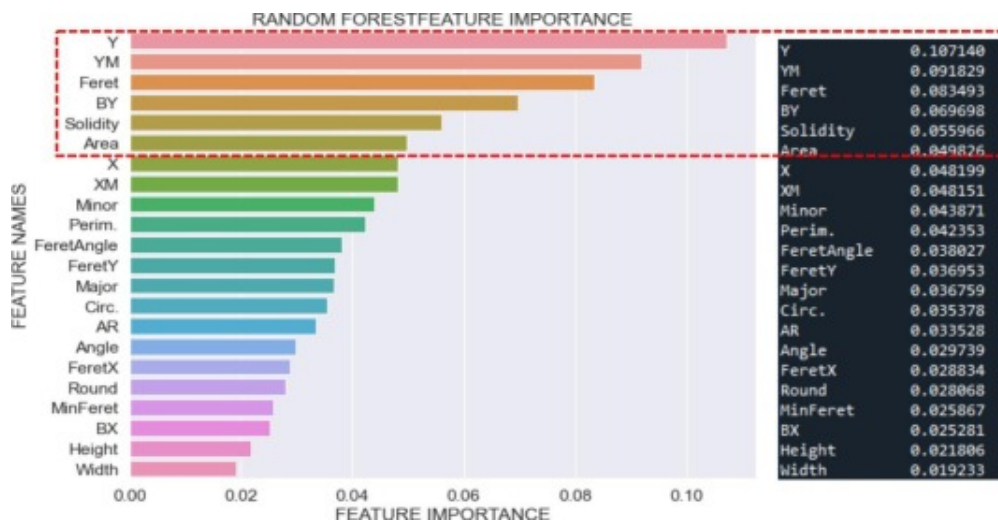
These features consistently ranked highest in terms of model reliance across multiple training iterations, indicating that they capture meaningful patterns associated with the pass/fail classification. The underlying hypothesis is that even in frames where the keyhole appears stable and no porosity is directly visible, there may be latent indicators, or transitional characteristics, which precede or follow defect formation. These hidden features, such as slight asymmetries or fluctuations in intensity, may reflect subtle process instabilities not immediately visible to the human eye but detectable through statistical learning. Features with lower importance had minimal impact on prediction accuracy when permuted, suggesting they were either redundant or weakly correlated with the true weld quality label. These results demonstrate that a data-driven machine learning model, when trained on image-derived features, can provide accurate, interpretable, and automated classification of weld quality. The use of Random Forest enabled both robust performance and transparent feature ranking, making it well-suited for industrial deployment in real-time weld inspection systems.

3.4.1. Permutation feature importance (PFI)

To identify the most relevant input features presented in [Table 3](#) for porosity classification, a Permutation Feature Importance (PFI) analysis was applied to the trained Random Forest model. PFI is a model-agnostic technique that quantifies each feature's contribution by measuring the drop in classification accuracy when its values are randomly shuffled. In this study, PFI was computed using a held-out 20% test set, with

importance scores representing the average reduction in predictive accuracy caused by permutation.

A total of 22 morphological and geometrical features were initially extracted from the binary weld cross-section images. These descriptors characterize multiple aspects of the weld nugget's geometry, including shape, size, symmetry, and boundary regularity, enabling a multi-dimensional evaluation of weld quality. The Random Forest feature importance ranking (Fig. 10) revealed that feature contributions were unevenly distributed, only a few features carried dominant predictive weight, while several others provided secondary or redundant information.



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Fig. 10. Random forest feature importance analysis.

As shown in Fig. 10, six features exhibited the highest importance scores and were therefore retained for model development: vertical centroid position (Y, 0.107), brightness-weighted center of mass (YM, 0.091), Feret's diameter (0.083), bounding box Y-coordinate (BY, 0.067), solidity (0.056), and area (0.049). These features collectively

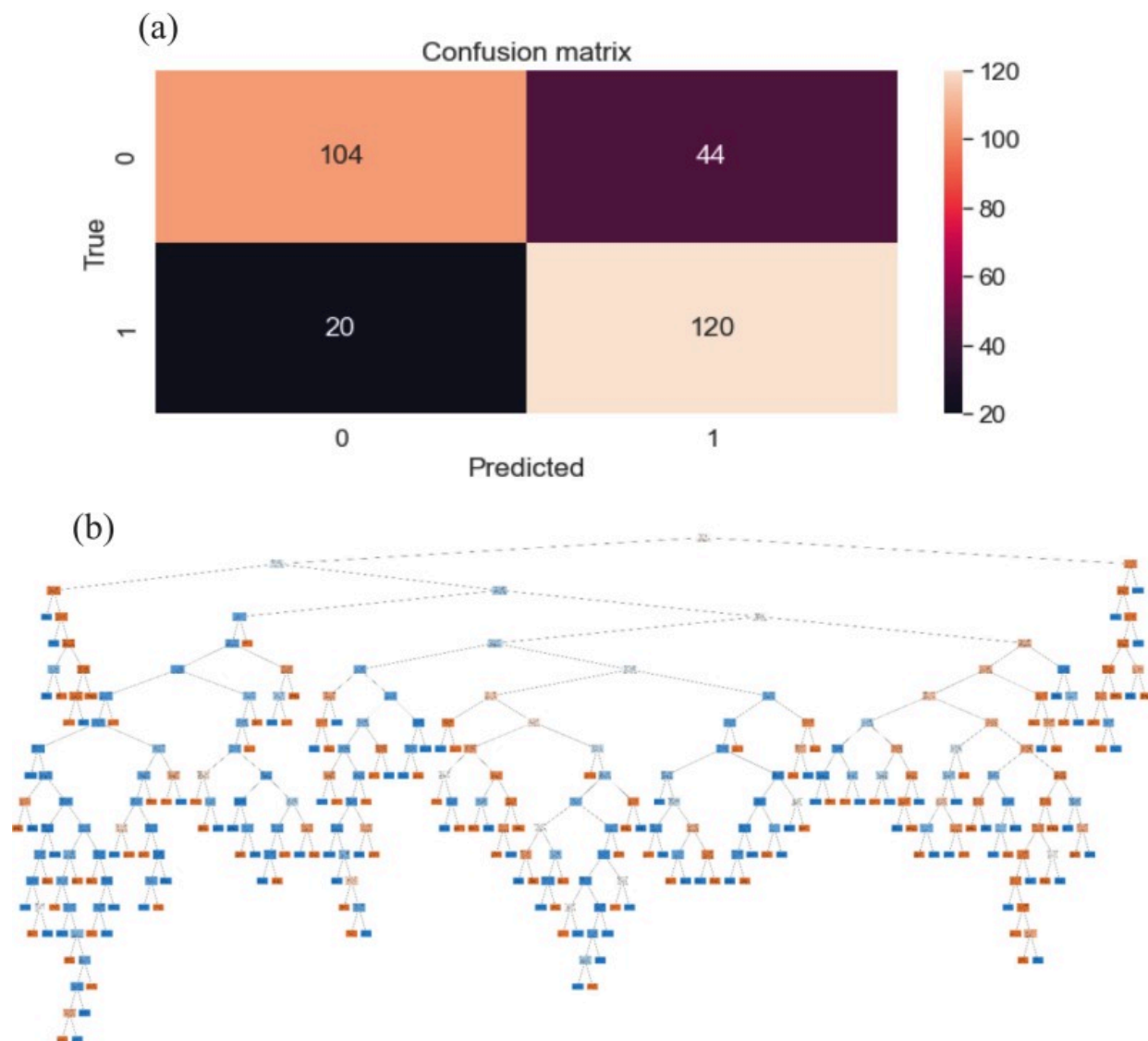
capture positional, morphological, and structural attributes of the weld nugget, offering the greatest discriminative capacity for porosity classification. The vertical centroid (Y) and intensity-weighted moment (YM) describe nugget displacement and asymmetry along the laser axis, which are strongly linked to instability and expulsion. Feret's diameter and solidity quantify geometric elongation and compactness, reflecting keyhole collapse or incomplete fusion, while area measures the overall molten pool extent and absorbed energy input. Together, these features form the primary predictive basis of the model.

Moderately important features (importance values between 0.03 and 0.05), such as the horizontal centroid (X, 0.048), X-moment (XM, 0.048), minor axis (0.048), perimeter (0.043), Feret angle (0.043), and major axis (0.037), contributed additional refinement by capturing secondary geometric effects including elongation, tilt, and surface irregularity. These features enhanced the model's sensitivity to subtle morphological variations without substantially altering predictive accuracy.

Low-ranked features (<0.03), such as aspect ratio (AR, 0.033), ellipse rotation angle (0.033), FeretX (0.029), roundness (0.028), minimum Feret diameter (0.026), bounding box X-coordinate (BX, 0.025), height (0.022), and width (0.019), exhibited limited unique contribution and were excluded from the final model due to their strong correlation with dominant size or shape descriptors.

Collectively, this hierarchical ranking indicates that the model primarily learns from morphological asymmetry and size heterogeneity, consistent with the physical behavior of spot welding, where nugget displacement, expulsion, and incomplete fusion directly alter projected area and centroid location. Among the 22 extracted image features, centroid-based (Y, YM) and shape-compactness descriptors (Feret, Solidity, Area) demonstrated the highest discriminative power (>0.05), underscoring their dominant role in predicting weld quality. The exclusion of redundant and low-impact features streamlined the model while preserving accuracy and interpretability, improving both computational efficiency and physical transparency.

To evaluate the classification performance of the trained Random Forest model, a confusion matrix and a representative decision tree structure were generated, as shown in Fig. 11. The confusion matrix (Fig. 11a) summarizes the model's predictions on the test set, which comprised 20% of the total dataset. The matrix reveals that out of 124 actual defective (Fail=0) samples, 104 were correctly predicted, while 44 were misclassified. For the non-defective (Pass=1) class, 120 out of 140 samples were correctly identified, with only 20 false negatives. These results correspond to an overall classification accuracy of 78%, with a precision of 0.84 for the Pass class and 0.73 for the Fail class. The recall for the Fail class was notably high at 0.86, indicating strong sensitivity in detecting porosity-prone weld segments. In addition to performance metrics, the internal logic of the Random Forest model is illustrated through a single decision tree (Fig. 11b). The visualization reflects the hierarchical decision-making process based on the selected geometric and spatial features of the keyhole. The tree nodes indicate feature-based splitting thresholds that segment the data into increasingly homogeneous subsets, contributing to the model's prediction capability. While Random Forest operates through ensemble averaging over many such trees, this visualized tree offers interpretability and transparency regarding how feature conditions lead to classification outcomes.

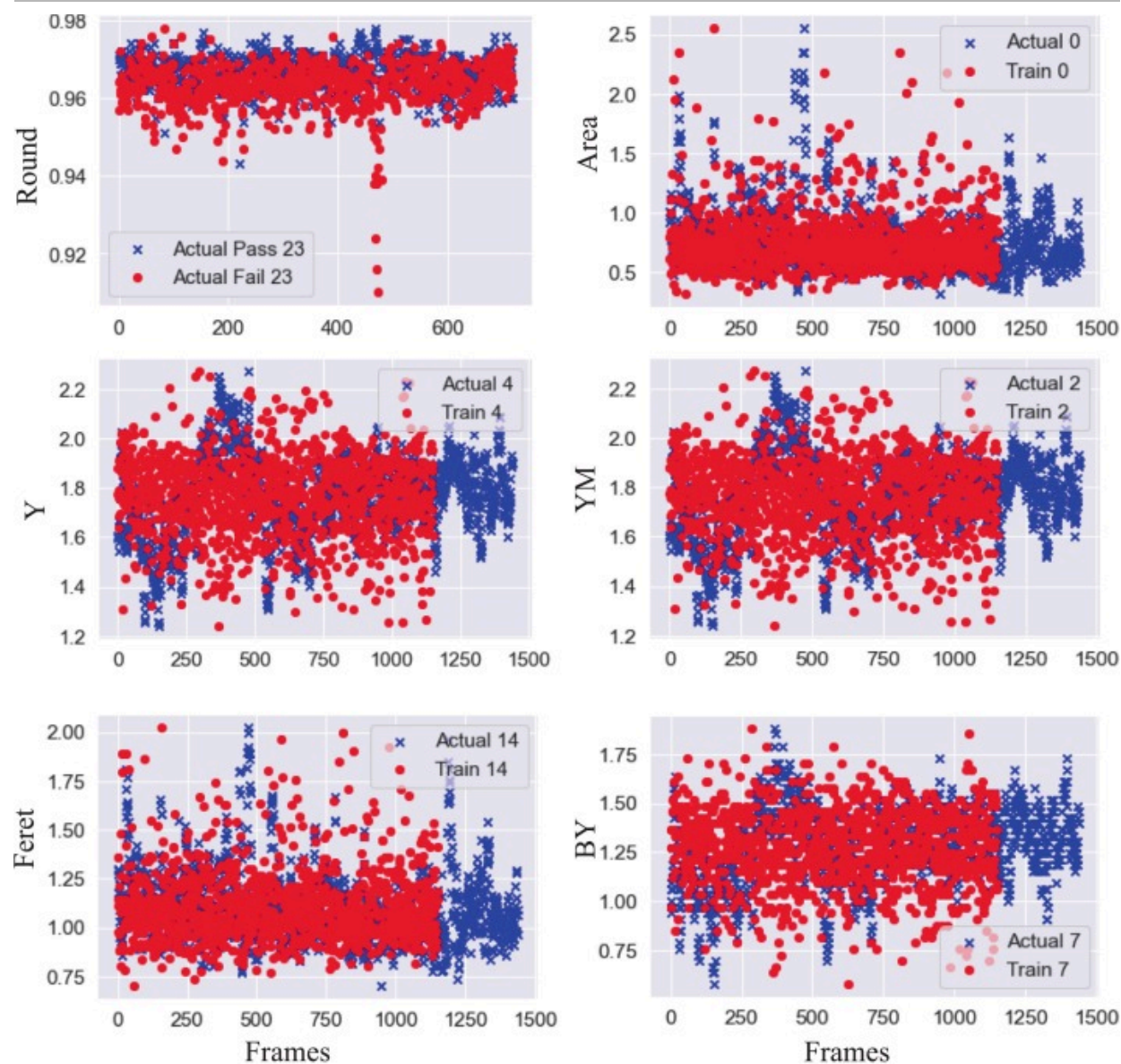


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Fig. 11. Random Forest Prediction. a) Confusion matrix Test size 20 % and b) Random Forest decision tree.

To further evaluate the model's predictive reliability and understand how well it captures feature behavior, an actual-versus-predicted feature plot was generated, as shown in Fig. 12. In this plot, each data point represents a frame from the test set. The x-axis indicates the frame index, and the y-axis represents the magnitude of a specific extracted geometric feature. Blue markers denote the actual feature values, while red markers correspond to the predictions generated by the Random Forest model. A close overlap between the two sets of points indicates high predictive accuracy across the weld sequence. The relatively tight clustering of predicted values around the actual values suggests that the model effectively captures the temporal dynamics of these features. Occasional dispersion likely reflects frames where feature values fluctuate more erratically or are influenced by overlapping class characteristics. This combination of statistical and visual evaluation demonstrates that the Random Forest model, trained on geometrical and thermal image features, is capable of both accurate porosity prediction and transparent interpretation. The use of Permutation Feature Importance enables not only the quantification of feature relevance but also supports deeper insights into the underlying physical phenomena captured during the welding process. Together, these results substantiate the potential for integrating machine learning-based analytics into real-time quality monitoring systems, where both precision and interpretability are essential for industrial implementation.



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Fig. 12. Extracted geometric features: Actual Vs Predictive Model.

3.4.2. Model performance report

Table 4 presents the performance metrics of the classification model, including precision, recall, and F1-score for both Pass and Fail classes. Precision is defined as the ratio of true positives to the sum of true and false positives, measuring the model's accuracy in predicting a specific class. Recall represents the ability to correctly identify all actual instances of a class (true positives divided by actual positives). The F1-score is the harmonic mean of precision and recall, providing a balanced measure of performance. These metrics offer a comprehensive evaluation of model behavior across imbalanced classes and are supported by the overall accuracy and macro- and weighted averages. The results of the random forest classification model indicate an overall accuracy of 78%, with precision of 0.84 for the "Pass" class and 0.73 for the "Fail" class. The recall score for the "Pass" class is 0.70, while the recall score for the "Fail" class is 0.86. The F1-scores for the "Pass" and "Fail" classes are 0.76 and 0.79, respectively. The precision score for the "Pass" class indicates that when the model predicts a "Pass" result, it is correct 84% of the time. Similarly, the precision score for the "Fail" class indicates that when the model predicts a "Fail" result, it is correct 73% of the time. The recall score for the "Pass" class suggests that the model correctly identified 70% of the "Pass" results, while the recall score for the "Fail" class indicates that the model correctly identified 86% of the "Fail" results. The F1-score provides a balance between precision and recall, with the score for the "Pass" class being 0.76 and for the "Fail" class being 0.79. Overall, the results suggest that the random forest classification model is effective in predicting the occurrence of porosity in keyhole laser welding of overlap aluminum laser welding, with an accuracy of 78%. This indicates that the model is able to correctly classify 78% of the samples in the dataset. The precision, recall, and F1-score metrics suggest that the model performs better at predicting the "Pass" class than the "Fail" class, which may indicate a class imbalance in the dataset. It is important to note that the accuracy of the model could be further improved by increasing the size of the dataset, as well as by fine-tuning the model hyperparameters. Additionally, further analysis of the features used in the model could provide insights into the most crucial factors contributing to the occurrence of porosity in laser welding. In conclusion, the random forest classification model is a promising

approach for predicting the occurrence of porosity in laser welding and could be useful in quality control and process optimization in industrial settings. However, further research is needed to optimize the model and to validate its effectiveness in real-world applications.

Table 4. Performance metric of model.

	Precision	Recall	F1-score	Support
Pass	0.84	0.70	0.76	148
Fail	0.73	0.86	0.79	140
Accuracy	NA	NA	0.78	288
Macro avg	0.79	0.78	0.78	288
Weighted avg	0.79	0.78	0.78	288

NA: Not applicable.

4. Conclusion

This study introduces a data-driven approach for real-time porosity monitoring in overlap laser welding of AA6061 aluminum alloy, integrating high-speed imaging, geometric feature extraction, and machine learning. A synchronized top-view high-speed camera operating at 720 fps was used to capture weld-pool dynamics, from which six key geometric and spatial features were extracted to characterize keyhole behavior. The Random Forest classifier trained on these features achieved a porosity classification accuracy of approximately 80%, with predictions validated against post-process X-ray imaging. The results confirmed a strong correlation between geometric instability, particularly fluctuations in centroid position, Feret's diameter, and roundness, and porosity formation. Segment-based evaluation across 6mm weld regions enabled

detection of porosity-prone zones with 85% accuracy, leading to a demonstrated 80% reduction in undetected defects. The proposed framework offers a scalable foundation for intelligent, in-line quality assurance in laser welding. Nevertheless, the current implementation is limited by its frame-wise independence; the model does not incorporate temporal information or contextual memory, which constrains its ability to capture dynamic weld evolution. Future research will focus on developing memory-enabled architectures, such as Long Short-Term Memory (LSTM) or Transformer-based models, to model temporal dependencies in keyhole behavior. This advancement will be critical for enabling real-time adaptive control and achieving closed-loop process regulation, ultimately supporting the deployment of autonomous welding systems in smart manufacturing environments.

Notably in this study, all laser parameters, including power, travel speed, and wobble settings (0.5mm amplitude, 200Hz frequency), were kept constant to isolate porosity-relevant effects. Consequently, the influence of different oscillation trajectories and amplitudes on keyhole behavior and porosity formation was not explored. This limitation is acknowledged, and future work will address this gap through expanded datasets and controlled variation of oscillation parameters. Advancing toward memory-enabled models will further support real-time adaptive control in smart laser welding systems.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used AI tools to assist with language correction, readability improvement, and overall clarity of the manuscript. After using these tools, the authors carefully reviewed and edited the content as necessary and take full responsibility for the final version of the manuscript.

CRediT authorship contribution statement

Ahmad Aminzadeh: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sasan Sattarpanah Karganroudi:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Data curation, Conceptualization. **Narges Omid:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis. **Nouredine Barka:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis. **Abderrazak El Ouafi:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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